

PRATEEK SINGH

B.Tech CSE, IIIT Guwahati (2023–2027) | Ps332927@gmail.com | 9473668934 | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)
GSSoC 2026 Project Admin | GSSoC 2025 Contributor | Hacktoberfest 2025 | CodeSocial 2025

PROFESSIONAL SUMMARY

B.Tech Computer Science student at IIIT Guwahati with hands-on experience across full-stack web development, applied AI/ML, and workflow automation. Skilled in building end-to-end products with Next.js, FastAPI, and modern NLP/RAG pipelines, including an enterprise feedback-analytics platform and a conditional GAN for image colorization. Proven track record of collaborating in remote, cross-functional environments as a GSSoC Project Admin, open-source contributor, and research collaborator with the University of Alaska Anchorage.

EDUCATION

Indian Institute of Information Technology, Guwahati

2023 – 2027

B.Tech in Computer Science & Engineering

TECHNICAL SKILLS

Programming Languages: Java, Python, C++, JavaScript, TypeScript, HTML, CSS

Frontend: React.js, Next.js 14, TailwindCSS, Vite, Context API, CSS Modules

Backend: Node.js, Express.js, FastAPI, REST APIs, Servlet/JSP, JDBC

Databases: MySQL, MongoDB, Mongoose, SQLite, Firebase (Auth & Firestore), ChromaDB

Machine Learning & NLP: PyTorch, Scikit-learn, GANs (Pix2Pix, U-Net, PatchGAN), BERTopic, NLTK VADER, Sentence-Transformers, LangChain (RAG), NumPy, Pandas, Matplotlib, Seaborn, Albumentations

Tools & Platforms: Git, GitHub, VS Code, Linux, Jupyter Notebook, n8n, Ollama, Vercel

Core CS: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS, Computer Architecture, Discrete Mathematics, Secure Coding (OWASP)

PROJECTS

BizInsight AI — Enterprise Customer Feedback Analytics Platform — [GitHub](#)

2026

Next.js 14 • FastAPI • BERTopic • LangChain • ChromaDB • NLTK VADER • Sentence-Transformers

- Migrated a legacy monolithic Streamlit application to a modern microservices architecture with a Next.js 14 frontend and a FastAPI backend.
- Built automated sentiment analysis (NLTK VADER) and unsupervised topic clustering (BERTopic, HDBSCAN, UMAP, Sentence-Transformers) that groups negative reviews into business-relevant categories such as Payment, Delivery, and Product Quality.
- Developed a Retrieval-Augmented Generation chatbot (LangChain, ChromaDB, OpenRouter Gemini) that answers business queries strictly from uploaded review data to prevent hallucination, plus a standalone public RAG sandbox requiring no authentication.
- Implemented structured alerting for risk levels and negative-review spikes to support proactive business monitoring.

Automated Documentation Agent — [GitHub](#)

2026

n8n • Groq AI • Webhook Triggers • REST APIs

- Built a webhook-triggered workflow automation agent in n8n that uses Groq AI to auto-generate codebase documentation on trigger.
- Demonstrated agentic automation by chaining AI inference, webhook handling, and data transformation without any custom backend code.

Sketch-to-Colorization using Conditional GAN — [GitHub](#)

2025

Python • PyTorch • U-Net • PatchGAN • Albumentations • Jupyter Notebook

- Built a conditional GAN (Pix2Pix-based) in PyTorch to automatically colorize black-and-white anime-style sketches, using a U-Net generator with skip connections to preserve fine sketch detail.
- Implemented a PatchGAN discriminator that evaluates local image patches instead of the whole image, producing sharper, more realistic textures in generated output.
- Trained and validated the model end-to-end on the Anime Sketch Colorization Pair dataset, with the full data loading, training, and inference pipeline in a single reproducible notebook.

CodeCapsule — Offline AI Coding Assistant (VS Code Extension) — [GitHub](#)

2024

JavaScript • VS Code Extension API • Ollama • Local LLMs

- Developed a VS Code extension that runs entirely offline using local LLMs via Ollama, removing dependency on external APIs or an internet connection.
- Implemented in-editor code explanation, bug detection, refactoring suggestions, and code generation using the VS Code Extension API.

EXPERIENCE & OPEN SOURCE

Project Admin — GSSoC 2026 (GirlScript Summer of Code)

2026

- Manage contributor onboarding, PR reviews, and issue triaging for an open-source project under GSSoC 2026, maintaining code quality standards and overall project health.
- Mentor incoming contributors and ensure timely merges, supporting a smooth and active contribution pipeline.

Research Collaborator — KathiraveluLab, University of Alaska Anchorage

2025 – 2026

- Contributed to GUARDA (Gateway for Uniform Access to Remote Data & Analytics), a federated database query platform, under Dr. Pradeeban Kathiravelu's research group, applying secure coding practices and structured issue tracking while collaborating remotely across time zones.
- Received an official reference letter from Dr. Kathiravelu (Assistant Professor of CS, UAA) recognizing contributions to research and open-source collaboration.

Open Source Contributor — GSSoC 2025, Hacktoberfest 2025, CodeSocial 2025

2025

- Merged multiple PRs across repositories (bug fixes, cleanup, docs, features) following OWASP guidelines and real-world Git workflows (branching, rebasing, CI checks).